Lab Sheet: 2

Table Structures:

CLASS

D (NUMBER) and Name (VARCHAR).

The initial records in the table are (1, 'Bravo'), (2, 'Alex'), and (4, 'Cheng').

EMPLOYEES

Employee_Id (NUMBER), First_Name (VARCHAR), Last_Name (VARCHAR), Email (VARCHAR), Phone_Number (VARCHAR), Hire_Date (DATE), Job_Id (VARCHAR), Salary (NUMBER), Commission_Pct (NUMBER), Manager_Id (NUMBER), and Department_Id (NUMBER).

DDL Questions (Data Definition Language):

1. How do you create a table named CLASS with appropriate data types?

```
MySQL 8.0 Command Line Cli

Enter password: ****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 13
Server version: 8.0.35 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE SchoolDB;
Query OK, 1 row affected (0.01 sec)

mysql> USE SchoolDB;
Database changed
mysql> CREATE TABLE CLASS (
    ->     ID INT,
    ->     Name VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.08 sec)

mysql> INSERT INTO CLASS VALUES (1,'Bravo');
Query OK, 1 row affected (0.03 sec)

mysql> INSERT INTO CLASS VALUES (2,'Alex');
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO CLASS VALUES (4,'Cheng');
Query OK, 1 row affected (0.01 sec)

mysql>
```

2. How do you create a table named EMPLOYEES with the specified attributes?

```
mysql> CREATE TABLE EMPLOYEES (
    ->     Employee_Id INT,
    ->     First_Name VARCHAR(50),
    ->     Last_Name VARCHAR(50),
    ->     Email VARCHAR(100),
    ->     Phone_Number VARCHAR(20),
    ->     Hire_Date DATE,
    ->     Job_Id VARCHAR(50),
    ->     Salary INT,
    ->     Commission_Pct DECIMAL(5,2),
    ->     Manager_Id INT,
    ->     Department_Id INT
    -> );
Query OK, 0 rows affected (0.04 sec)
```

3. How do you create a table named Client_master with the specified attributes?

```
mysql> CREATE TABLE Client_master (
    ->     ClientNo VARCHAR(10),
    ->     Name VARCHAR(50),
    ->     Address VARCHAR(100),
    ->     City VARCHAR(50),
    ->     State VARCHAR(50),
    ->     Pincode INT,
    ->     Bal_due INT
    -> );
Query OK, 0 rows affected (0.04 sec)
```

4. What SQL statement would you use to rename the table Client_master to Client12?

```
mysql> RENAME TABLE Client_master TO Client12;
Query OK, 0 rows affected (0.04 sec)
```

5. How do you display the bal_due column heading as “BALANCE”?

```
mysql> SELECT Bal_due AS BALANCE FROM Client_master;
+---------+
| BALANCE |
+---------+
|    6000 |
|    4000 |
|    8000 |
|    3000 |
|    9000 |
+---------+
5 rows in set (0.00 sec)
```

DML Questions (Data Manipulation Language):

Part A: Transactions and Savepoints (CLASS Table)

Give the output of the following SQL script:

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO CLASS VALUES (5,'Rahul');
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> UPDATE CLASS SET Name='Abhijeet' WHERE ID=5;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SAVEPOINT A;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO CLASS VALUES (6,'Chris');
Query OK, 1 row affected (0.00 sec)

mysql> SAVEPOINT B;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO CLASS VALUES (7,'Bravo');
Query OK, 1 row affected (0.00 sec)

mysql> SAVEPOINT C;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> SELECT * FROM CLASS;
+------+----------+
| ID   | Name     |
+------+----------+
|    1 | Bravo    |
|    2 | Alex     |
|    4 | Cheng    |
|    5 | Abhijeet |
|    6 | Chris    |
|    7 | Bravo    |
+------+----------+
6 rows in set (0.00 sec)

mysql>
mysql> ROLLBACK TO B;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> SELECT * FROM CLASS;
+------+----------+
| ID   | Name     |
+------+----------+
|    1 | Bravo    |
|    2 | Alex     |
|    4 | Cheng    |
|    5 | Abhijeet |
|    6 | Chris    |
+------+----------+
5 rows in set (0.00 sec)

mysql>
mysql> ROLLBACK TO A;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)
```

Part B: Retrieving Records (EMPLOYEES Table)

1. Find out the employee id, names, and salaries of all the employees.

```
mysql> SELECT Employee_Id, First_Name, Last_Name, Salary
    -> FROM EMPLOYEES;
+-------------+------------+-----------+--------+
| Employee_Id | First_Name | Last_Name | Salary |
+-------------+------------+-----------+--------+
|         101 | John       | Austin    |   5000 |
|         102 | Neha       | Sharma    |   4500 |
|         103 | Amit       | Verma     |   6000 |
|         104 | Riya       | Austin    |   4800 |
|         105 | Karan      | Mehta     |   7000 |
+-------------+------------+-----------+--------+
5 rows in set (0.00 sec)
```

2. List out the employees who work under manager 100.

```
mysql> SELECT *
    -> FROM EMPLOYEES
    -> WHERE Manager_Id = 100;
+-------------+------------+----------+----------------+--------------+------------+---------+--------+----------------+------------+---------------+
| Employee_Id | First_Name | Last_Name | Email          | Phone_Number | Hire_Date  | Job_Id  | Salary | Commission_Pct | Manager_Id | Department_Id |
+-------------+------------+----------+----------------+--------------+------------+---------+--------+----------------+------------+---------------+
|         101 | John       | Austin   | john@gmail.com | 9876543210   | 2020-01-10 | IT_PROG |   5000 |           0.10 |        100 |            60 |
|         103 | Amit       | Verma    | amit@gmail.com | 9876543212   | 2018-07-20 | SA_REP  |   6000 |           0.15 |        100 |            80 |
|         105 | Karan      | Mehta    | karan@gmail.com | 9876543214   | 2017-09-18 | MK_REP  |   7000 |           0.12 |        100 |            90 |
+-------------+------------+----------+----------------+--------------+------------+---------+--------+----------------+------------+---------------+
3 rows in set (0.01 sec)
```

3. Find the names of the employees who have a salary greater than or equal to 4800.

```
mysql> SELECT First_Name, Last_Name
    -> FROM EMPLOYEES
    -> WHERE Salary >= 4800;
+------------+-----------+
| First_Name | Last_Name |
+------------+-----------+
| John       | Austin    |
| Amit       | Verma     |
| Riya       | Austin    |
| Karan      | Mehta     |
+------------+-----------+
4 rows in set (0.01 sec)
```

4. List out the employees whose last name is ‘AUSTIN’.

```
mysql> SELECT *
    -> FROM EMPLOYEES
    -> WHERE Last_Name = 'AUSTIN';
+-------------+------------+----------+----------------+--------------+------------+---------+--------+----------------+------------+---------------+
| Employee_Id | First_Name | Last_Name | Email          | Phone_Number | Hire_Date  | Job_Id  | Salary | Commission_Pct | Manager_Id | Department_Id |
+-------------+------------+----------+----------------+--------------+------------+---------+--------+----------------+------------+---------------+
|         101 | John       | Austin   | john@gmail.com | 9876543210   | 2020-01-10 | IT_PROG |   5000 |           0.10 |        100 |            60 |
|         104 | Riya       | Austin   | riya@gmail.com | 9876543213   | 2021-11-05 | IT_PROG |   4800 |           0.08 |        102 |            60 |
+-------------+------------+----------+----------------+--------------+------------+---------+--------+----------------+------------+---------------+
2 rows in set (0.01 sec)
```

5. Find the names of the employees who work in departments 60, 70, and 80.

```
mysql> SELECT First_Name, Last_Name
    -> FROM EMPLOYEES
    -> WHERE Department_Id IN (60,70,80);
+------------+-----------+
| First_Name | Last_Name |
+------------+-----------+
| John       | Austin    |
| Neha       | Sharma    |
| Amit       | Verma     |
| Riya       | Austin    |
+------------+-----------+
4 rows in set (0.01 sec)
```

6. Display the unique Manager_Id.

```
mysql> SELECT DISTINCT Manager_Id
    -> FROM EMPLOYEES;
+------------+
| Manager_Id |
+------------+
|        100 |
|        101 |
|        102 |
+------------+
3 rows in set (0.02 sec)
```

PART C – Updating Records (Client_master Table)

1. Insert five records into the Client_master table.

```
mysql> INSERT INTO Client_master VALUES
    -> ('C123','Ravi','MG Road','Delhi','Delhi',110001,6000),
    -> ('C124','Anita','Park Street','Kolkata','WB',700001,4000),
    -> ('C125','Rahul','Brigade Road','Bangalore','KA',560001,8000),
    -> ('C126','Neha','Marine Drive','Mumbai','MH',400001,3000),
    -> ('C127','Amit','Sector 15','Noida','UP',201301,9000);
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

2. Find the names of clients whose bal_due > 5000.

```
mysql> SELECT Name
    -> FROM Client_master
    -> WHERE Bal_due > 5000;
+-------+
| Name  |
+-------+
| Ravi  |
| Rahul |
| Amit  |
| Ravi  |
| Rahul |
| Amit  |
+-------+
6 rows in set (0.00 sec)
```

3. Change the bal_due of ClientNo "C123" to Rs. 5100.

```
mysql> UPDATE Client_master
    -> SET Bal_due = 5100
    -> WHERE ClientNo = 'C123';
Query OK, 2 rows affected (0.01 sec)
Rows matched: 2  Changed: 2  Warnings: 0
```